

Myungjoon Kim

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Education

- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
Ph.D., Materials Science and Engineering *Sep 2018 – Feb 2023*
Thesis: Inverse design of free-form metasurfaces: from adjoint-based optimization to physics-informed deep learning (*Advisor: Prof. Jonghwa Shin*)
- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
M.S., Materials Science and Engineering *Mar 2015 – Feb 2017*
- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
B.S., Materials Science and Engineering, Computer Science (Double major) *Feb 2008 – Feb 2015*
*incl. military service as a Korean Augmentation To the United States Army (KATUSA), 2010–2012

Research Interests

AI for physics · Differentiable simulation & inverse design · Computational photonics

Research Experience

- **Cornell University** **Dec 2024 –**
Schmidt AI in Science Postdoc (PI: Prof. Frank Wise) **Ithaca, USA**
 - Developing differentiable simulation and machine learning methods for ultrafast light control in nonlinear multimode photonics
- **Korea Advanced Institute of Science and Technology (KAIST)** **Apr 2023 – Nov 2024**
Postdoctoral Researcher (PI: Prof. Jonghwa Shin) **Daejeon, Korea**
 - Led development of machine learning and optimization methods for photonic simulation and design
- **Computational Science Research Center, KIST** **Jan 2018 – Aug 2018**
Research Intern (PI: Dr. Donghun Kim) **Seoul, Korea**
 - Developed a slab-graph convolutional neural network (SGCNN), extending CGCNN to surface structures, for predicting adsorption energies of catalytic systems

Selected Projects

- **Differentiable simulation & inverse design**
 - Implemented differentiable simulators for nonlinear multimode pulse propagation (GMMNLSE and (3+1)D GNLSE equations) in PyTorch, enabling gradient-based inverse design of spatiotemporal light control—optimizing input fields to target spatial, temporal, and spectral output properties [[GitHub](#)]
 - Developed adjoint-based optimization frameworks for free-form metasurfaces, designing phase masks for 3D photolithography—experimentally realized [[Science Advances, 2022](#)]

- Devised a Consensus-ADMM field optimization algorithm, achieving higher-performing structures than topology optimization across a range of photonic design problems [\[GitHub\]](#)
- **Machine learning for physics**
 - Built equivariant neural surrogate solvers for Maxwell's equations, encoding symmetries into the architecture to match the accuracy of non-equivariant baselines with over 30% less training data [\[GitHub\]](#)
 - Developing mode-informed neural operators in the guided eigenmode basis of multimode fibers, learning spatiotemporal pulse propagation from both simulation and experimental data with over $500\times$ faster inference than numerical solvers, and applying them to inverse design of ultrafast light control [\[SPIE Optics+Photonics, 2026; manuscript in prep.\]](#)
 - Built an LLM-based workflow for nanophotonic inverse design—generating simulation code via prompt engineering and fine-tuning models to solve inverse design tasks directly [\[Nanophotonics, 2025\]](#)

Publications

('†' indicates equal contribution)

- V Sharma, N Bender, H Haig, **M Kim**, DN Christodoulides, FW Wise*, "Observation of stability of Gaussian beams and off-axis beam-cleaning in graded-index rods", [Optics Express, 33, 48130 \(2025\)](#)
- JR Lee†, **M Kim**†, SH Moon, DS Choi, J Shin, IS Kang, TJ Je*, E Jeon*, "Manufacturing of a bitmap-type lightweight microwave absorber metasurface", [International Journal of Precision Engineering and Manufacturing, 26, 3461 \(2025\)](#)
- **M Kim**†, H Park†, J Shin*, "Nanophotonic device design based on large language models: multilayer and metasurface examples", [Nanophotonics, 14, 1273 \(2025\)](#)
- D Lee, M Lee, B Yerenzhep, **M Kim**, H Hugonnet, S Jeon, J Shin, Y Park, "Fast free-form phase mask design for three-dimensional photolithography using convergent Born series", [ACS Photonics, 12, 610 \(2024\)](#)
- M Cho, J Jung, **M Kim**, J Lee, S Min, J Hong, S Lee, M Heo, JU Kim, I-S Joe, J Shin*, "Color arrestor pixels for high-fidelity, high-sensitivity imaging sensors", [Nanophotonics, 13, 2971 \(2024\)](#)
- **M Kim**†, N Kim†, J Shin*, "Realization of all two-dimensional Bravais lattices with metasurface-based interference lithography", [Nanophotonics, 13, 1467 \(2024\)](#)
- N Kim, **M Kim**, J Jung, T Chang, S Jeon, J Shin*, "Highly angle-sensitive and efficient optical metasurfaces with broken mirror symmetry", [Nanophotonics, 12, 2347 \(2023\)](#)
- A Baucour, **M Kim**, J Shin*, "Data-driven concurrent nanostructure optimization based on conditional generative adversarial networks", [Nanophotonics, 11, 2865 \(2022\)](#)
- SH Nam†, **M Kim**†, N Kim†, D Cho, M Choi, JH Park, J Shin*, S Jeon*, "Photolithographic realization of target nanostructures in 3D space by inverse design of phase modulation", [Science Advances, 8, abm6310 \(2022\)](#)
- S Hong*, CH Liow, JM Yuk, HR Byon, Y Yang, EA Cho, J Yeom, G Park, H Kang, S Kim, Y Shim, M Na, C Jeong, G Hwang, H Kim, H Kim, S Eom, S Cho, H Jun, Y Lee, A Baucour, K Bang, **M Kim**, S Yun, J Ryu, Y Han, A Jetybayeva, P-P Choi, J C Agar, S V Kalinin, P W Voorhees, P Littlewood, H M Lee, "Reducing time to discovery: materials and molecular modeling, imaging, informatics, and integration", [ACS Nano, 15, 3971 \(2021\)](#)
- N Kim, T Chang, **M Kim**, M Heo, A Baucour, J Jung, J Shin*, "Spectrally sharp metasurfaces for wide-angle high extinction of green lasers", [Optics Express, 28, 22121 \(2020\)](#)

- **M Kim**[†], BC Yeot[†], Y Park, HM Lee, SS Han*, D Kim*, "Artificial Intelligence to Accelerate the Discovery of N₂ Electroreduction Catalysts", [Chemistry of Materials](#), 32, 709 (2020)
- H Kim, **M Kim**, T Chang, A Baucour, S Jeon, N Kim, H-J Choi, H Lee, and J Shin*, "Bright and vivid plasmonic color filters having dual resonance modes with proper orthogonality", [Optics Express](#), 26, 27403 (2018)
- J Oh, J Kim, **M Kim**, W Choi, S Lee, U Lee*, "Understanding Mobile Document Capture and Correcting Orientation Errors", [International Journal of Human-Computer Studies](#), 104, 64 (2017)

Conferences and Presentations

- **M Kim**, V Sharma, N Bender, FW Wise, "Inverse design for spatiotemporal control of ultrafast light in optical fiber via physics-informed machine learning", SPIE Optics + Photonics (2026)
- (Invited Talk) **M Kim**, "Optimizing Photonic and Material Design with Differentiable Programming", Guest Lecture, AI for Materials, Cornell University (2025)
- (Invited Talk) **M Kim**, J Park, J Shin, "Toward accurate neural Maxwell solvers for inverse design of metasurfaces", KPS Spring meeting (2024)
- **M Kim**, J Park, J Shin, "Efficient Fourier neural operators by group convolution and channel shuffling", ICLR Workshop on AI4DifferentialEquations in Science (2024)
- **M Kim**, N Kim, J Shin, "Concurrent Inverse Design of Structured Light and Metasurface for Nanopatterning Process", Frontiers in Optics (2022)

Skills

- **Languages:** Python, MATLAB
- **ML frameworks:** PyTorch, JAX
- **Simulations:** FDTD (Tidy3D, Lumerical, Meep), FDFD, RCWA, DDA, nonlinear pulse propagation (GMMNLSE, (3+1)D GNLSE)

Academic Service

- **Reviewer:** ICLR, NeurIPS